

Introduction

ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing) measures genome-wide chromatin accessibility by tagging accessible DNA regions with a hyperactive Tn5 transposase. Peaks in the resulting alignment density correspond to open chromatin — regulatory elements such as promoters and enhancers — and differential accessibility analysis between conditions reveals cell-state-specific regulatory changes. ATAC-seq has largely replaced DNase-seq for accessibility profiling because of its lower input requirement (10⁴ cells vs millions) and simpler protocol.

Prerequisites

A working understanding of read alignment, BAM files, peak-calling concepts (typically MACS2), and regulatory genomics (promoters, enhancers, transcription factor binding).

Theory

The standard ATAC-seq pipeline:

1. **QC:** TSS enrichment score, fragment-length distribution with characteristic nucleosomal peaks (mono-, di-, tri-nucleosomal at approximately 150, 300, 450 bp).
2. **Duplicate and mitochondrial-read removal:** typical mitochondrial fraction is 20–50 % of reads and must be removed before peak calling.
3. **Peak calling:** MACS2 with Tn5-specific flags (`--nomodel --shift -75 --extsize 150`) treats each end of the fragment as a Tn5 cut site.
4. **Consensus peak set:** combine peaks across replicates and conditions for joint quantitative analysis.
5. **Differential accessibility:** DiffBind, edgeR, or DESeq2 on peak-level read counts.
6. **Peak annotation:** ChIPseeker assigns peaks to nearest gene and annotates promoter vs distal categories.

Assumptions

Tn5 tagmentation is approximately uniform across accessible regions; sequencing depth is sufficient (> 50 million reads per sample is typical); mitochondrial reads have been removed before peak calling.

R Implementation

```
library(ATACseqQC); library(ChIPseeker)
library(TxDb.Hsapiens.UCSC.hg19.knownGene)

# Fragment-length diagnostic from an example ATAC bam shipped with ATACseqQC
bam <- system.file("extdata", "GL1.bam", package = "ATACseqQC")
# fragSize <- fragSizeDist(bam, bamFiles.labels = "example")

# Peak annotation example with a toy peak GRanges
set.seed(2026)
peaks <- GRanges(
  seqnames = rep("chr1", 50),
  ranges = IRanges(start = sort(sample(1e7:1.05e7, 50)),
    width = 500)
)
anno <- annotatePeak(peaks,
  TxDb = TxDb.Hsapiens.UCSC.hg19.knownGene,
  level = "gene",
  verbose = FALSE)
as.data.frame(anno)[1:5, c("seqnames", "start", "annotation", "geneId")]
```

Output & Results

`ATACseqQC::fragSizeDist()` produces the diagnostic fragment-size distribution; visible peaks at the nucleosomal lengths confirm successful tagmentation. `ChIPseeker::annotatePeak()` assigns each peak to the nearest gene and categorises it as promoter, intronic, intergenic, etc.

Interpretation

A reporting sentence: “ATAC-seq on CD8 and regulatory T cells (n = 4 per group) identified 25{,}412 reproducible peaks at $\text{IDR} < 0.05$; ChIPseeker annotation showed 35 % promoter-proximal and 65 % distal-regulatory locations. Differential accessibility analysis identified 1{,}240 peaks at $\text{FDR} < 0.05$, enriched for AP-1 family motifs (Jun, Fos).” Always report TSS enrichment, peak counts, and the consensus peak strategy.

Practical Tips

- Remove duplicates and mitochondrial reads with `samtools markdup` and chromosome filtering before peak calling; mitochondrial reads dominate ATAC libraries and confound peak calling if retained.
- Check TSS enrichment score (computed by `ATACseqQC` or by `chrom.sizes`-aware tools); scores above 7 are typical for fresh tissue, scores above 4 are acceptable for frozen samples.
- For differential accessibility, define a consensus peak set across replicates (DiffBind-style) and quantify all samples on the same regions; calling peaks separately per sample produces unstable comparisons.
- Motif enrichment via `chromVAR` provides per-sample transcription-factor activity scores; pair with differential accessibility to identify TFs driving the observed changes.
- For single-cell ATAC-seq (scATAC-seq), use `Signac` (Seurat-flavoured) or `ArchR`; the workflow differs substantially from bulk ATAC.
- Use BlackList regions (ENCODE) to exclude artefact-prone genomic regions before downstream analysis.

R Packages Used

`ATACseqQC` for fragment-size distribution and TSS enrichment QC; `ChIPseeker` for peak annotation; `DiffBind` for differential-binding analysis; `chromVAR` for motif accessibility scores; `Signac` and `ArchR` for single-cell ATAC-seq; `Rsubread` and `GenomicAlignments` for upstream alignment and quantification.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with `DESeq2` / `edgeR`
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with `Seurat`